



Figure 4: Ganglia Web UI

You should see the Ganglia Web interface displaying the metrics of the Ganglia instance itself.

Installing the Host sFlow agent

With your primary Ganglia monitoring system set up, all you need to do now is install and configure the Host sFlow agent on the remaining cloud instances that you wish to monitor.

Depending on the device you want to monitor, you can download the appropriate agent from http://sourceforge.net/projects/host-sflow/files/REL-1_22/

Since the instance we are using here is a simple Linux instance, let's use the standard Host sFlow agent RPM download. Once downloaded, run the RPM as shown:

```
# wget http://sourceforge.net/projects/host-sflow/files/REL-1_22/hsflowd-1.22-1.x86_64.rpm
# rpm -ivh hsflowd-1.22-1.x86_64.rpm
```

With the RPM installed, all you need to do is edit the sFlow configuration file and point the agent to the Ganglia *gmond* instance:

```
# vi /etc/hsflowd.conf

# Add the following code in your hsflowd.conf file
sflow{
    DNSSD = off
    polling = 20
    sampling = 512
    collector{
        ip = <your_ganglia_server_IP_Address>
        udpport = 6343
    }
}
```

After editing the configuration file, you will need to restart the Host sFlow agent:

```
# service hsflowd start
```

After the sFlow agent is started, uncomment the following entry in the *gmond* configuration file (*/etc/gmond.conf*) in your Ganglia monitoring system instance:

```
# vi /etc/gmond.conf
/* sflow_channel */
udp_recv_channel {
    port = 6343
}
```

The Ganglia charts will now start displaying the performance metrics collected using the sFlow agent. Enabling sFlow monitoring on each instance in such environments can thus provide a simple, efficient and highly scalable solution for monitoring performance and workloads in clouds. 

References

- [1] <http://ganglia.sourceforge.net/>
- [2] <http://www.sflow.org/sFlowOverview.pdf>
- [3] <http://velocityconf.com/velocity2012/public/schedule/detail/23487>
- [4] <http://blog.sflow.com/2010/10/ganglia.html>

By: Yohan Wadia

The author is a senior software engineer and is a part of the Cloud Services team (Research and Innovation) at IGATE. An avid blogger and technologist, he loves exploring all emerging technologies and trends in the IT industry. Blog: www.yoyoclouds.com.